IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Received 10 May 2024, accepted 19 May 2024, date of publication 23 May 2024, date of current version 31 May 2024.

Digital Object Identifier 10.1109/ACCESS.2024.3404406

RESEARCH ARTICLE

Optimizing Urban-Scale Evacuation Strategies Through Disaster Victim Aggregation Modification

REZA ASRIANDI EKAPUTRA[1,2], SEONG-HOON KEE[1], AND JURNG-JAE YEE[1]
[1]Department of ICT Integrated Safe Ocean Smart Cities Engineering, Dong-A University, Busan 49315, South Korea
[2]Civil Engineering Department, Institut Teknologi Sumatera, Lampung Selatan, Lampung 35365, Indonesia

Corresponding author: Jurng-Jae Yee (jjyee@dau.ac.kr)

This work was supported by the Basic Science Research Program through the National Research Foundation (NRF), funded by Korean Ministry of Education under Grant 2016R1A6A1A03012812.

ABSTRACT As urban areas continue progressing into more complex development, urban areas become more vulnerable to disaster. Therefore, there is growing interest in investigating disaster response processes considering population distribution variation within urban regions. This study explores the unsupervised machine learning methods, K-means, DBSCAN, and introduces a novel Ensemble Clustering, which is compared to the established grid-based method as a framework for aggregating disaster victims in determining evacuation strategies. The Ensemble Clustering method combines the K-Means clustering strengths in dividing disaster victim distribution into globular cluster distribution. DBSCAN supports it as a robust outlier identification for disaggregating the irrelevant member as an individual. Subsequently, the clustering effectivity is evaluated by simulating the evacuation scenarios in the evacuation simulation, and the performance is compared to the traditional grid-based method. Combining the machine learning method and the evacuation simulation can be a framework for the decision-making process during a disaster. The results show that the Ensemble Clustering outperforms the traditional grid method, ensuring a faster completion time.

INDEX TERMS City-scale evacuation, evacuation strategy, ensemble clustering, machine learning, pedestrian evacuation.

I. INTRODUCTION

The frequency and intensity of floods have significantly increased, leading to massive infrastructure damage, economic loss, and social consequences [1]. Floods may occur due to a variety of factors, such as high-intensity rainfall, rapid snow melt, or dam failure [2], [3], [4]. One of the notable floods was a flood in Pakistan in 2010 that affected over 1,700 people deaths and evacuated around 20 million people [5]. Another catastrophic event was the New Orleans flood caused by Hurricane Katrina in 2005, which devastated the East Gulf area in the United States and caused extensive infrastructure damage [6]. Recognizing the potential damages from disaster, proactive effort becomes vital to prevent losses during disaster situations. Therefore, developing an evacuation strategy framework is essential to ensure community safety and reduce potential loss when facing disaster [7]. However, evacuation is a complex system involving many stakeholders and resources. Evacuation is classified as a complex system because the evacuation process component is non-linearly correlated and interrelated [8]. This complexity leads to challenges in increasing the decision-making time in determining the evacuation strategies [9], [10]. Furthermore, the literature review revealed that evacuation decisions are important to increase the evacuation time efficiency, and the most successful evacuation occurs when evacuees have sufficient time throughout the evacuation process [10], [11]. Consequently, studies in developing evacuation frameworks that aim to minimize evacuation time are important to improve the decision-making process during a disaster.

Different approaches are developed to determine the framework for reducing evacuation time. The commonly

The associate editor coordinating the review of this manuscript and approving it for publication was Qiang Li.

© 2024 The Authors. This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

used approach involves modifying the supplies side of the evacuation process. Several existing studies discuss private-vehicle based evacuation and propose strategies that eliminate potential traffic congestion by implementing contraflow [11], [12]. This technique can increase the traffic network capacity and reduce evacuation time. However, contraflow development may not always apply to any disaster location. Therefore, an alternative solution that can be implemented is developing an evacuation rerouting strategy. Rerouting redirects evacuation movement to different road sections in the related road network. The rerouting process ensures an equilibrium traffic load distribution and avoids traffic jams on particular road sections [14], [15]. Additionally, a traffic management strategy combined with dynamic evacuation enables an adaptive strategy that follows the demand fluctuation [16].

In specific scenarios, private vehicle evacuation may not be practical due to limited regional accessibility during a natural disaster situation. Consequently, walking-based or transit vehicle-based evacuation is also developed as an alternative evacuation strategy. Research in pedestrian evacuation mainly focuses on investigating the individual response and analyzing the human interaction process during the evacuation. The evacuee interaction is strongly related to the victim's location, emotional state, personal relationships, and social situation [17], [18], [19]. Moreover, multiple studies on pedestrian evacuation examine how the surrounding environment affects the disaster victim's perception and changes the evacuation patterns on various evacuation scales (small-scale and urban scale).

In the context of urban-scale evacuations, it is acknowledged that a transit-based evacuation can significantly reduce demand on road networks [20]. Transit-based evacuation has been extensively discussed, with the most popular topic being a round-trip evacuation model during disasters, which aims to determine the optimal pickup point and route in the evacuation process [21], [22], [23]. Several objective functions are utilized, such as waiting time and pickup location, which are considered crucial due to their impact on the accessibility of disaster victims to the initial stage of the evacuation process [24]. Additionally, for a more straightforward analysis, some studies assume that each evacuee already waits in the evacuation pickup-point location [25]. In addition, the study of bus evacuation also includes dynamic demand analysis, which plays a crucial role in facilitating the evacuation demand aspect, particularly when dealing with massive-scale disasters that require an effective evacuation process [26]. Large-scale evacuation concepts must be evaluated through a comprehensive and detailed procedure that divides the evacuation analysis into smaller stages. For example, in a limited human resources case during an evacuation, the evacuation is separated into different stages, estimating the number of people involved in the evacuation process [27]. Another discussion is categorizing the stages according to the progressive development of the disaster, making the evacuation decision more relevant to the situation in the surrounding area [28].

An alternative approach to formulating urban-scale evacuation strategies involves disaster victim aggregation. While most studies have focused on the supply side of evacuations, addressing factors that involve route optimization, determining shelter location, or evaluating multimodal evacuation, there is a growing interest in exploring the demand side analysis for developing an evacuation strategy [29], [30].

Furthermore, studying the demand side of the evacuation process is crucial as it serves as an initial stage for analyzing the supply needs for evacuation. This finding suggests that an aggregate level of analysis is important to be applied when developing evacuation strategies. The disaster victim aggregation method can significantly enhance the evacuation process efficiency by enabling stakeholders to make informed decisions based on these evacuation groups [31], [32].

There are two commonly used aggregation methods. The first method involves victim aggregation based on regional boundaries [33], [34]. This aggregation supports more effective coordination for the evacuation process, and the evacuation decision can be gradually delivered from higher levels of government to the lower administrative level. However, the administrative-based aggregation found challenges in the scalability due to the administrative area being a fixed arrangement and when facing a cross-administrative boundary disaster situation. Grid-based aggregation, the second method, divides disaster victims into equal-sized grids. This approach assumes that each disaster victim within the same grid forms an evacuation group [35]. This method plays a crucial role in improving computational processes during evacuation analysis. However, the grid approach does not guarantee accurate clustering of all data, particularly in areas with external geographical characteristics [36], [37].

Therefore, several studies have focused on modifying the grid-based approach to improve the disaster victim aggregation process. An example is a grid-density-based grouping for crowd evacuation, which generates a novel grouping algorithm that can identify different density locations [38]. A non-uniform binary grid approach was introduced as an iterative method to determine a smaller cluster-grid group following a binary pattern [39]. However, it is essential to note that these studies are conducted on a local scale and assume movement based on cells, which does not accurately represent a city-scale evacuation. Moreover, other studies have utilized machine learning algorithms to improve disaster victim aggregation, although these methods require further improvement when facing a city-scale evacuation [40]. A city-scale evacuation needs an actual road network and involves interaction between disaster victims. Therefore, disaster victim aggregation can be combined with evacuation simulation to contribute to an integrated robust framework for developing evacuation strategies.

In developing evacuation strategies, researchers have used evacuation simulations to generate evacuation plans [41]. The

utilization of evacuation simulations is essential to create effective evacuation strategies because these simulations accurately represent the actual process of evacuation by studying the interactions between different components of the evacuation [42]. Therefore, evacuation simulations have the potential to assist in planning large-scale evacuations that involve complex scenarios, including variations in scale, algorithm, and location. These simulations are essential in evacuation strategy development that evaluates the interaction between individuals, individuals and vehicles, and vehicles with vehicles during the evacuation process. By using these simulations, researchers can analyze and evaluate the evacuation component movement progress during an evacuation. Evacuation computer simulation utilizes traffic simulation applications in developing evacuation strategies. This application enables a comprehensive assessment of people and vehicles during evacuation.

Additionally, computer simulations also facilitate analysis scalability in evacuation planning. Ejercito et al. conducted a systematic review that compared various simulation modeling platforms, including MATSim, SUMO, Aimsun, PTV Vissim, and GAMA [43]. Evacuation simulation has an advantage in continuously analyzing the evacuation process with high precision [44]. SUMO simulation offers a significant advantage due to its free license that allows SUMO to be used in the traffic simulation process. This licensing regulation enables SUMO to have connectivity with various supporting applications, such as visualization and analysis. SUMO microsimulation becomes more advantageous than other traffic simulations because it can be executed on its graphical user interface (GUI) and integrated with the Python programming library [45]. Furthermore, supporting modules are readily available on the online program storage platform. In SUMO simulation, SUMO provides two types of pedestrian movement options that can be utilized based on specific needs.

The non-interacting model, which serves as the fundamental model in SUMO, is ideal for model movements that do not require considering the interaction between individuals in the simulation. However, a stripping model is more suitable for evacuation modeling because it can consider the short-range interaction of persons on the road network. Moreover, SUMO has an advanced intermodality model as its input data components can be constructed in a more flexible design form [46]. Constructing a traffic simulation in SUMO involves several stages, including the establishment of a transportation network, the definition of traffic conditions, the selection of a routing algorithm, and the execution of a simulation. Each stage is stored in separate standardized data that can be connected. The transport network data is stored in "*net.xml*" data converted from primary OSM data containing geometry, length, and road network position. The demand of the network is stored in the *"pessenger.xml"* data that saves the origin, destination, evacuation mode, and movement stages of the people on the network. The *"net.xml"* and *"pessenger.xml"* *are* connected by the *"sumo.cfg"* file as configuration for the simulation.

This study evaluates the disaster victim aggregation process for a city-scale evacuation in a flood-induced disaster. Three algorithms, K-means, DBSCAN, and a novel Ensemble Clustering method, are utilized to handle various population distributions and improve the precision of the disaster victim aggregation process. The methodology, hypothetically, can develop a clustering process by evaluating distance data points to the centroid and sequentially separating the diverging data within clusters as independent entities. Combining K-Means clustering and DBSCAN can improve the simulation initialization process and lead to a more relevant decision-making process during emergencies.

This article is structured as follows. **Section I** comprehensively explains the background, motivation, and relevant literature, providing a solid foundation for understanding the research topic and its associated theories. Following is **Section II** outlines the proposed methodology for this research. This section elaborates on the data collection and analysis techniques employed in the study. **Section III** presents the research findings, which include the results and discussion on the impact of varying cluster numbers for each method, the influence of different data distributions, a comparison of the best-performing clusters for each technique, and the enhancement of the distribution model through the combination of k-means and DBSCAN. **Section IV** explains the study's conclusion and proposes suggestions for future research.

II. METHODOLOGY

Figure 1 presents a comprehensive framework that illustrates the overall process of disaster victim aggregation and simulation during flood-induced disasters. This research consists of three main stages: data input, analysis, and evacuation simulation. Different datasets are utilized to evaluate the effectiveness of the evacuation strategy. The first dataset consists of a flood risk map with a return period of 100 years, which takes a conservative approach to developing evacuation strategies. This conservative approach may lead to overestimating risk and potentially result in increased disaster preparedness allocations, but ensures a higher safety level. The second dataset involves building area data from the Open Street Map (OSM) to estimate the population.

The third dataset includes the location of emergency shelters provided by the National Disaster Management Institute (NDMI). The final data is the road network from the OSM that presents the raster data connecting points in the actual road network. Subsequently, the population data are aggregated using different clustering methods and then combined with the road network information to generate Origin-Destination (O-D) matrices as inputs for the simulation of evacuations. SUMO simulation generates an advanced modeling process that involves pedestrian evacuation dynamics and simulates it in an actual evacuation process. This step analytically evaluates the route choice

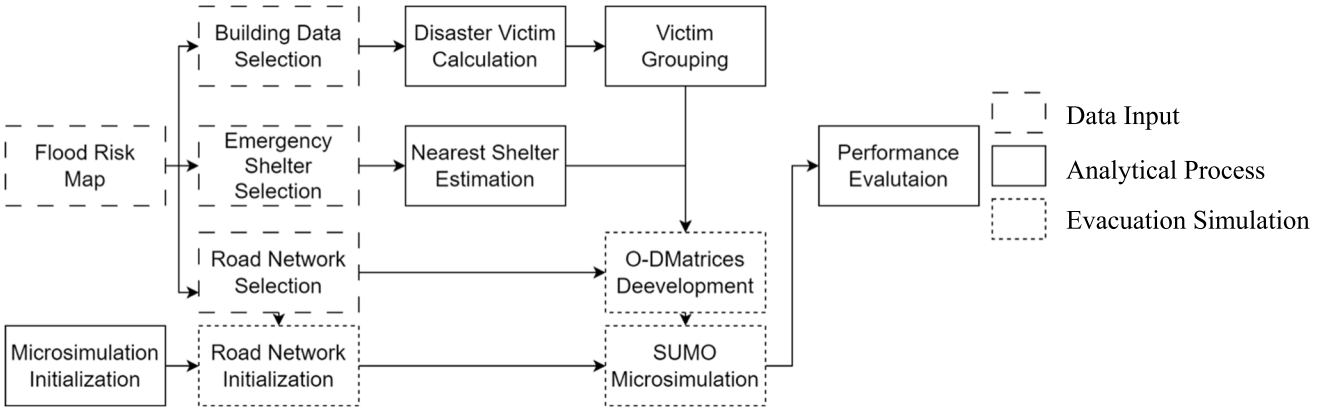


FIGURE 1. General flow chart of the study.

during the evacuation. It assesses the surrounding area at each step to find the moving pattern of each disaster victim in the evacuation process. The comprehensive framework contributes to the study in determining an evacuation strategy that aims to evacuate people to the emergency shelter effectively during a flood situation.

A. STUDY AREA

The Busan Metropolitan Area (**Figure 2**) stands as the second largest city in South Korea, spanning approximately 770 Km^2 [47]. This city is positioned on the south-eastern coastline of the Korean peninsula, and it borders Ulsan to the north and Gyeongsangnam-do to the west. As one of the most vital urban centers in South Korea, this city plays an important role in supporting economic and social stability. Additionally, it serves a wide range of economic activities, including logistic management, tourism activity, and residential areas. The Busan metropolitan area has experienced several devastating flood events and possible indications for future potential disaster occurrences. These flood events impact the entire Busan Metropolitan Area region, but the most significant incidents are predicted to occur along the Nakdong River catchment area, which is evaluated as the main case in this study (**Figure 2**, location A). The Nakdong River, which originates in the northern region and flows through the Busan Metropolitan Area is the longest in South Korea. The Nakdong River is a crucial source of clean water for the Busan area and contributes significantly to the local economy. However, the geographic location of the Nakdong River exposes the Busan Metropolitan Area to potential hazards and disasters [48].

Additionally, smaller-scale flood prone areas are observed in comparison to the main case study. This additional case is utilized for validating the proposed framework. The first location is the disaster-prone area surrounding the Central Canal in Dongrae-Gu, Busan, South Korea (**Figure 2**, location B) and the second location is located along the East Coast of Busan, (Jung-Gu and Seo-Gu) (**Figure 2**, location C). These locations are selected due to their unique location and population distribution characteristics. The Central Canal

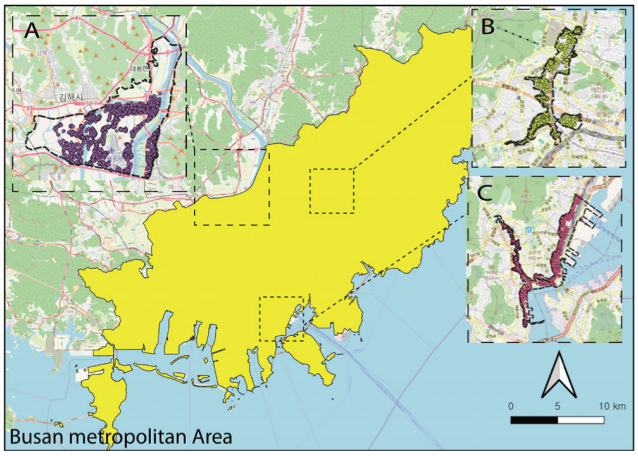


FIGURE 2. Busan metropolitan area, South Korea.

and East Coast disaster-prone areas are smaller than the Nakdong River-affected areas with a centralized population distribution. This variation makes this case study important to showcase the scalability of the proposed method. Moreover, the East Coast region presents additional complexity, with various commercial areas interspersed within the population centers. This complexity provides valuable insights into the efficacy of the proposed aggregation method.

B. DATA COLLECTION

1) POPULATION DISTRIBUTION DEFINITION

Population databases are essential for developing evacuation strategies. The population databases provide the spatial location of disaster victims in the x- and y- axis locations. This study utilizes the *Universal Transverse Mercator*(UTM) coordinate transformation to ensure consistency throughout the analysis process. This study population dataset focuses on the individuals residing within multiple flood-prone areas in the Busan Metropolitan Area, located inside the Nakdong River catchment area, Busan East Coast, and Busan Central Canal. In this boundary, it is estimated that approximately more than 12,000 individuals are prone to disaster and must be evacuated to a safer area in a flood situation based on the

disaster risk map. The population estimation is based on a calculation approach using Geographic Information Systems (GIS) building data and a housing statistics dataset to estimate the population affected by the disaster [49]. The GIS building area method requires input from the building statistical database obtained from the OSM, which includes information such as *x-axis* coordinates, *y-axis* coordinates, utilization type, and building area size. The database is then sorted to include only buildings with residential utilization types and the population within the disaster-affected area is calculated based on this information. For the population dataset involved in this research, it is assumed that all residents in the database will evacuate from their residences during a disaster. All the evacuees are already aware of the necessary actions and evacuation process. Moreover, this study does not include visiting families or tourists in disaster-affected areas.

2) EMERGENCY SHELTER LOCATION

The process of relocating disaster victims to emergency shelter locations is called an evacuation. These emergency shelters are provided by stakeholders as temporary residences during disasters. Developed regions, such as the Busan Metropolitan Area, typically maintain a government database of selected shelter locations for each region. The emergency shelter location includes various types of public spaces: school, parking lot, bank, subway station, and government office [50]. When evaluating evacuation locations, the government typically considers factors such as accessibility and vulnerability. The main focus of this parameter is to ensure that evacuees can be immediately relocated to the shelter. According to the Busan government database, over 4,000 emergency shelter locations exist. During evacuation These locations are plotted on the Quantum GIS (QGIS) application to support spatial analysis and visualization [51]. Some adjustments are needed to use the dataset during the evacuation process, with the most important modification being the conversion of the emergency shelter coordinate system from the Central Korean Belt (CKB) system to the UTM system.

3) TRANSPORTATION NETWORK DATASET

Disaster victims are relocated during disaster situations using the regional road network. Therefore, defining the available road network for evacuation is crucial when developing strategies. In SUMO Simulation, the road network is represented as vector data with multiple features [52].

The first feature is the node, representing a road section's starting and ending points. The second part is the edge, which connects two nodes and can include one or more geometries. The final component is the network management system, which contains essential information about the road network, such as geometry (number of lanes and width), allowed and disallowed traffic modes, traffic light schedule, and traffic direction information. This research employs data from OSM acquired through the "*netconvert*" module in SUMO Simulation.



FIGURE 3. Road network data set from open street map (OSM) database.

C. ANALYSIS PROCESS

1) CLUSTERING ALGORITHM

Clustering is the main component that contributes to the disaster victim aggregation analysis. Various clustering approaches apply to diverse applications. This research employs several clustering methods to evaluate the distribution of disaster victims and generate accurate disaster victim aggregation. The first approach utilizes K-means clustering as part of partition-based clustering that groups data based on the proximity of each data to a cluster center [53]. K-means clustering is advantageous in assessing globular data by optimizing the objective function, *e.g.,* the sum of distances, to determine the optimal cluster member for each cluster center [54]. **Algorithm 1** presents the pseudocode of the K-means clustering. However, in the case of a non-globular data distribution, K-means clustering encounters challenges in identifying an accurate data distribution for its corresponding members within the dataset, resulting in sub-optimal clustering outcomes.

Algorithm 1 Pseudo Code

Input:
K (Cluster Number Value)
$D = \{x_1, x_2, \ldots, x_n\}$ (Data Component),
Output:
$C = \{c_1, c_2, \ldots, c_K\}$ (Set of K centroids),
$S = \{S_1, S_2, \ldots, S_K\}$ (Group S in K clusters)

1: Initialize K and Select C (randomly) for each data D
2: Repeat:
3: $\forall \{x\} \in D$, do:
4: $\delta(x) = \min\{c \in C\}$, |x - c| (vector distance x and c)
5: Assign x to cluster S {with min $\delta(x)$}
6: $\forall S_j \in K$, do:
7: $c_j = (1/|Sj|) \sum\{|x| \in S_j\}$ (calculate new centroid)
8: Update $c_j = \{c \in K\}$
9: Until C converges

Therefore, the DBSCAN method is utilized for this research by evaluating data distribution based on distance threshold (e) and minimum points (n) between neighboring data. The cluster that does not meet the threshold is considered as noise. The DBSCAN approach is effective for analyzing non-globular-shaped clusters and identifying group outliers [55]. **Algorithm 2** presents the DBSCAN Pseudo Code.

DBSCAN effectively manages large datasets, especially di distribution with different densities, and can separate noise for data that is not connected to any neighbor. However, DBSCAN is disadvantageous when dealing with data distributions with similar densities and generating a large cluster size. Combining K-means and DBSCAN can potentially improve the development of disaster victim aggregation in determining evacuation plans during city-scale and non-globular data distribution. As a result, this study introduces an innovative Ensemble Clustering technique that combines the strengths of K-means and DBSCAN algorithms. This approach is designed to effectively aggregate data on disaster victims and facilitate the development of more efficient evacuation strategies. This combination follows a process that establishes K-means as an initial step in constructing a globular group. Subsequently, DBSCAN focuses on segregating data points that can be distributed as a smaller group and evaluating the unconnected to the group as individuals that address the surrounding environmental constraints. The K-means clustering is an iterative process to assign disaster victims into clusters. It determines the suitable cluster based on each data's distance to the centroid. This study utilizes the *x-axis* coordinate, *y-axis* coordinate, and *r-variable*, representing the data point's administrative location. The objective function J can be formulated in (1).

$$J(x, y, r) = min \sum_{j=1}^{k} \sum_{i=1}^{n} w_{ij}[\left(x_i^{(j)} - c_{jx}\right)^2 + \left(y_i^{(j)} - c_{jy}\right)^2 + (r_i^{(j)} - c_{jr})^2] \quad (1)$$

where $J(x,y,r)$ is the objective function of the research for cluster number K and n number data points, each data point has three main features: X_i, Y_i, and r_i that represent the *x-axis* coordinate, *y-axis* coordinate, and administrative location of the reviewed location, (c_{jx}, c_{jy}, and c_{jr}) are the centroids feature for each cluster and w_{ij} is a value indicator (binary value) for X_i, Y_i, and r_i with $w_{ij} = 1$ if the set of data is assigned to cluster j, and 0 otherwise. This algorithm iterates until convergence, which is typically determined by observing no significant change in the centroids' coordinates across consecutive iterations. After the initial clustering process, the dataset undergoes further refinement using DBSCAN to generate sub-clusters.

DBSCAN scans each data point's neighborhood within a specified distance threshold (e). A new cluster is formed for a cluster with a member count equal to or surpassing the minimum point threshold (*minPts*). This cluster then expands by assimilating neighboring points that meet the criteria,

Algorithm 2 DBSCAN Clustering Pseudo Code

Input:
e (epsilon-max distance to neighboring data),
minPts (minimum number of points)
$D = \{x_1, x_2, \ldots, x_n\}$ (Dataset component)
Output:
$N = \{n_1, n_2, \ldots, n_j\}$(Noise data list)
$C = \{c_1, c_2, \ldots, c_i\}$(Cluster data list),
Procedure in neighboring:
1: $\forall \{x\} \in D$, mark x as unvisited
2: While unvisited points $\in D > 0$:
3: Select an unvisited point $x \in D$
4: Mark x as $x*(visited)$
5: Set Neighbor $x_n = \{x \in D \mid ||q - p|| \leq e\}$
6: If $|\Sigma x_n| < minPts$:
7: Mark x_n as NOISE
8: Else:
9: New Custer C_j
10: End

refining the clustering structure. The algorithm begins by marking all data points as unvisited and then examines each unvisited point iteratively. Then, for D the dataset of j cluster contains $\{p_{j1}, p_{j2}, \ldots, p_{jn}\}$ that present the point member of cluster j with d as the distance between each data. Firstly, establishing the centroid point by following (2).

$$N_{\in}\left(p_{jn}\right) = \{p_j \in d\left(p_{jn}\right) \leq \epsilon\} \wedge N_{\in}\left(p_{jn}\right) = \{p_j > \text{min}Pts\}$$
$$Else: \quad N_{\in}\left(p_{jn}\right) = Noise \quad (2)$$

where $N_{\in}\left(p_{jn}\right)$ represents the new subgroup within the jth cluster. Data distribution with the number of neighbors larger than or to *minPts*, generates a new cluster and is expanded by adding all reachable neighbors within an e distance for each point. On the contrary, p_{jn} that follow the $p_{jn} <$ min*Pts* criteria, assigned as an individual. **Algorithm 3** presents the pseudocode of the Ensemble Clustering method.

2) SIMULATION SETUP

This study focuses on a city-scale evacuation simulation to minimize the evacuation time for pedestrian-based mass evacuations using SUMO simulation. The simulation examines the evacuation process for three areas and includes more than 12,000 pedestrians who are expected to evacuate from their homes during flood situations to emergency shelters. Additionally, this study is limited to the pre-announced evacuation scenario, assuming that the flood has not yet developed and affected the disaster area. As a result, all evacuees move at their normal speed based on demographics. Furthermore, the simulation assumes that all road networks can serve the evacuation process without unexpected road closures or additional constraints that could arise during an unplanned emergency operation. The simulation was run ten times to

Algorithm 3 Ensemble Clustering (N, e, minPts)

Input:
K (Determined Cluster Number),
e(neighborhood control distance),
minPts (minimum points in the cluster)
$D = \{x_1, x_2, \ldots, x_n\}$ (Dataset component)

Output:
$N = \{n_1, n_2, \ldots, n_j\}$(Noise data list)
$C = \{c_1, c_2, \ldots, c_i\}$(Cluster data list),

1: Initialize K for data D
2: For:
3: ∀ point $x \in D$
4: Assign x to cluster C_i
5: $\delta(x) = \min\{c \in C\}$, |x - c| (vector distance x and c)
6: ∀ cluster C_i from the set of clusters C
7: ∀ *point* $p \in C_i$, mark p as unvisited
8: ∀ $S_j \in \{1, 2, \ldots, K\}$, do:
8: ∀ $\{x\} \in D$, mark p as unvisited
9: Mark x *as* $x*$(visited)
10: Set Neighbor $x_n = \{x \in D \mid ||q - p|| \leq e\}$
11: If $|\Sigma x_n| < minPts$:
12: Mark x_n as NOISE
13: Else:
14: New Custer C_j
15: End
16: ∀ centroid c ∈ C
17: Update c as means of Ci distance
18: Until centroids converge

Return the set of clusters and noise points.

obtain an average value, considering different aggregation processes and modifying three pedestrian attributes. There are three modified pedestrian attributes are as follows [56]:

Mingap: Desired distance between leader and follower (m).

DesiredMaxSpeed: Desired maximum velocity (m/s).

Impatience: Probability to cross at unprioritized crossings (0 to 1).

The evacuation speed of disaster victims constantly changes depending on the traffic situation and the evacuation process. To demonstrate the advantages of the proposed method, a grid-based aggregation is also simulated in this study. The hardware used for this experiment includes a PC with an Intel i7-9700 3 GHz CPU, 32 GB DDR4 RAM, Intel UHD Graphics 630 with 16 GB memory, and a SATA SSD storage with a reading speed of 550 MB/s.

III. RESULT AND DISCUSSION

A. ANALYSIS OF CLUSTERING RESULT

This study investigates four disaster victim aggregation algorithms: Grid clustering, K-means, DBSCAN, and an Ensemble Clustering method. Each algorithm utilizes different clustering parameters based on different types of algorithm processes. Grid-based clustering utilizes grid-size parameters (N) to generate different evacuation scenarios. The N values for the grid clustering are *700, 1100, 1500, 1900, and 2300*. These values are strategically chosen considering the main geographical constraint in the study area (Nakdong River), which varies in width from approximately 200 to 1200 meters. Therefore, the initial grid size was set at 700 meters, representing the mid-point of the width. Subsequent grids increase incrementally by 400 meters until reaching 2300 meters. This systematic increase allows for a comprehensive evaluation of the analytical area, ensuring that all variations in river width and related geographical features are accurately represented and analyzed within the disaster management framework. Another assumption, $x = 490 \times 10^3$ and $y = 3,890 \times 10^6$ is the reference point for the grid-based aggregation method, and this research does not change the reference point for different cluster sizes. K-means utilizes different parameters for the analytical process. The K parameter in K-means clustering represents the estimated total clusters from the clustering process. This study uses $K = 10, 20, 30 \ldots, 80$ to determine the best-performing value for the data distribution. On the other hand, DBSCAN assesses neighbor data by utilizing e values equal to *50, 100, 150, …, 450*. Ensemble Clustering utilizes the optimal parameters obtained from K-means and DBSCAN, assuming that the combination yields the most optimum results from the analysis.

Figure 4 illustrates the result of disaster victim aggregation for various clustering algorithms in the case study of Nakdong River Area. **Figure 4(a)** shows the grid-based disaster victim clustering. The grid-clustering divides the victim following a predetermined equivalent grid distribution. This approach offers an advantage in interpretability because this technique does not require any technical expertise to be understood by the stakeholder and is commonly employed in the decision-making process. However, this technique faces challenges when clustering data with geographical features in the surrounding area. The grid-based clustering cannot accurately define the cluster edge and differentiate distinct clusters [57]. This issue arises because this technique does not incorporate the inherent characteristics of data distribution and local variation [58]. Visual assessment of data group A in **Figure 4(a)** confirms the inaccurate grouping process. Rivers as a geographical boundary pass through this data group and make this group inaccurate. The lack of consideration for data distribution locality results in a bias in the clustering process. On the contrary, K-means organizes the data by considering its proximity to the centroid, making this method more adaptive to the data variability and scalability [59]. This clustering involves an iterative procedure to ascertain the optimal location of the centroid and assign cluster members to each cluster based on the distance to the centroid. The K-means clustering analysis does not directly incorporate geographical characteristics. However, geographical features

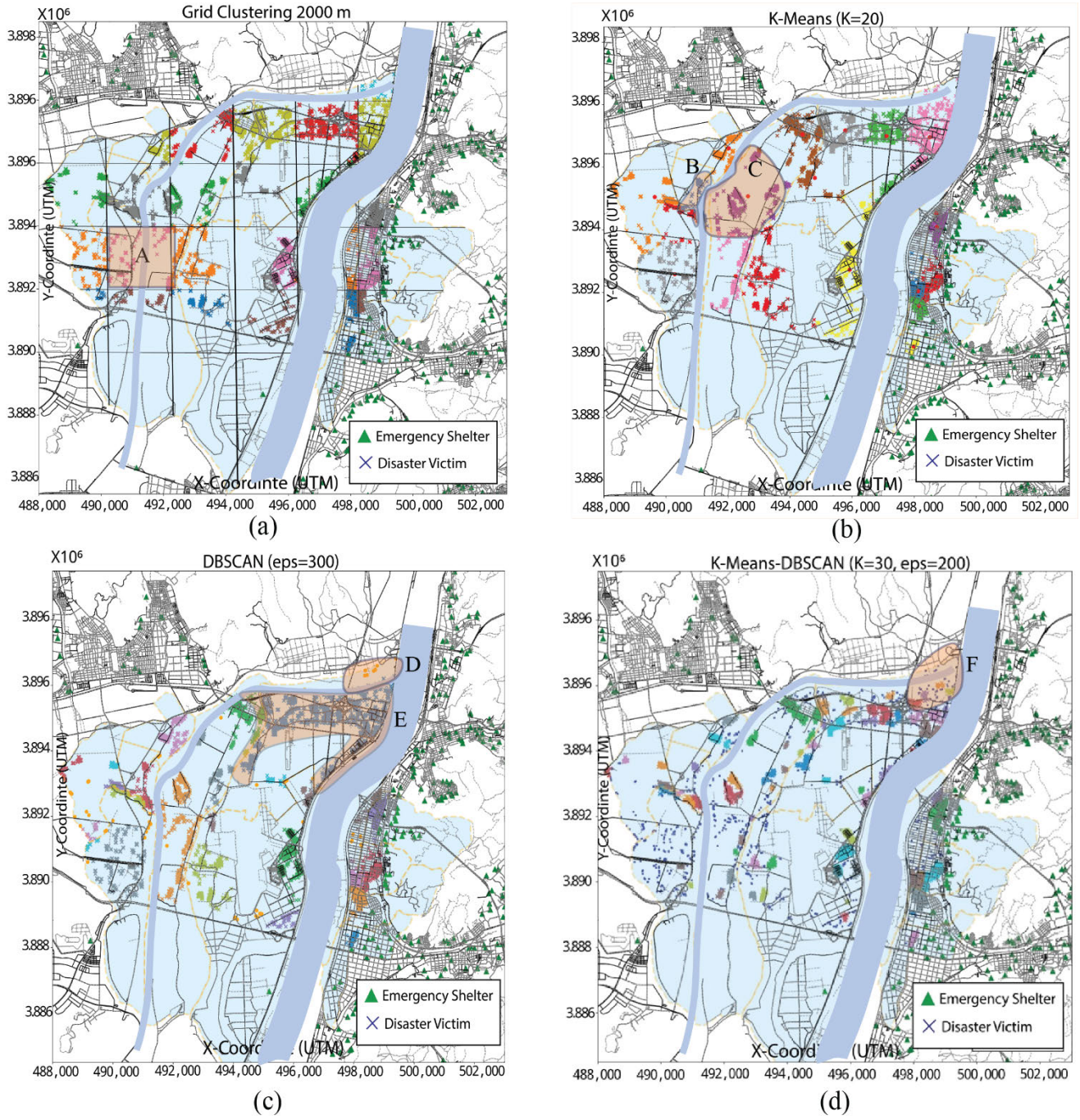


FIGURE 4. Distribution of Disaster Victim for Various Clustering Method (a) Grid Based (d = 2000m); (b) K-Means (K = 20); (c) DBSCAN (eps = 300m); (d) Combined (K = 30 and eps = 200).

incorporated into the data segregation can be recognized by the distance between the data points evaluated by the K-means clustering technique [60]. **Figure 4(b)** presents the K-means clustering results. This figure shows that most K-means clustering clusters can differentiate the cluster component divided by geographical features and improve the grid-based clustering method. For example, data group B and C in **Figure 4(b)** can be well separated and improves the grid-based clustering. However, determining the K value for K-means clustering is crucial during the analysis. Incorrect K value assignment can lead to failure in the K-means clustering process [61].

Furthermore, the DBSCAN clustering approach employs the parameters e and $minPt$ [62]. Utilizing these parameters eliminates the need for specifying the total cluster value (K) in the clustering process. Instead, the process identifies neighboring data with the maximum e value. The data group is classified as a cluster for neighboring data distance

less than e in the total cluster member exceeds the *minPt* threshold. Conversely, it is classified as noise. **Figure 4(c)** presents the DBSCAN clustering result example. Visual observation of this figure reveals that DBSCAN can cluster dataset by considering the local situation. This method can separate the incorporated data (data group D, **Figure 4(c)**) that K-means clustering cannot cover. However, this method generates a broader cluster due to the high-density data distribution (data group E, **Figure 4(c)**). This large cluster is irrelevant for disaster evacuation, leading to inaccurate evacuation site locations. Therefore, utilizing DBSCAN as the initial stage in the clustering method becomes irrelevant. Additionally, DBSCAN can generate multiple clusters with various sizes and exclude the outlier as an individual [63]. In this context, DBSCAN is suitable for the second stage of Ensemble Clustering. Ensemble clustering (**Figure 4(d)**) is a two-stage clustering method that utilizes the advantage of K-Means and DBSCAN. K-means works as the first stage of clustering by dividing the dataset into sets of globular clusters, and each cluster generates a smaller cluster to separate that might not be connected to the cluster. Data assigned as outliers are excluded from the aggregation and assigned as individuals for finding disaster victim locations. Data group F in Figure 4(d) illustrates the clustering transformation result. In K-means clustering, the data set is clustered as one cluster that assigns all the disaster victims to the same emergency shelter. On the contrary, after the Ensemble Clustering improvement, the group is divided into 3 smaller clusters and several outliers that support more flexible disaster victim distribution.

While **Figure 4** illustrates visual differences between various clustering methods, further evaluation using metrics is essential to ensure the method applicability and generalizability of this method [64]. Moreover, visual observation may fail to detect subtle differences, especially when the analysis involves smaller parameter values. Therefore, this research utilizes a *Silhouette* method in evaluating the data distribution. The *Silhouette* method is a metric to represent the intra-cluster cohesion and inter-cluster separation. A higher Silhouette value indicates the data is better clustered. The *Silhouette* method is presented in (3) [65].

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))} \tag{3}$$

The $a(i)$ variable in the (3) presents the mean distance separating the i data point to another point in a (cohesion). Similarly, the $b(i)$ variable denotes the shortest average distance between the a and b clusters (separation). **Figure 5** illustrates the *Silhouette Score* box plot of various clustering algorithms applied for the disaster victim location dataset. **Figure 5(a)** represents *Silhouette Score* for a grid-based clustering approach at different grid sizes. The *Silhouette Score* median generally shows a result between 0.3 and 0.6 for different grid sizes, indicating moderate cohesion and separation.

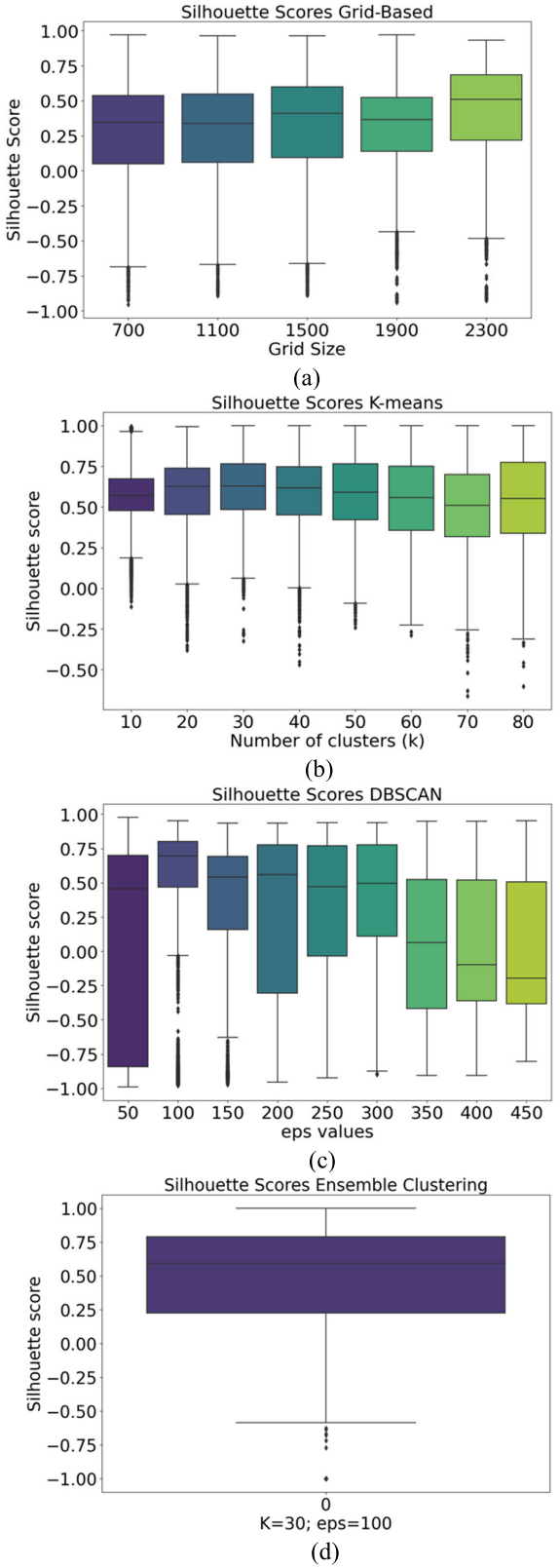


FIGURE 5. ***Silhouette*** **Scores Nakdong River (a) Grid-Based, (b) K-means, (c) DBSCAN, (d) Ensemble clustering.**

The highest median plot for grid-based clustering is associated with a grid size of 2300 m with 0.51 *Silhouette*

Score, and the lowest median plot is for a grid size of 700 m with 0.36 *Silhouette Score*. The observation for this image presents some data points that are not effectively clustered [66]. **Figure 5(b)** presents the *Silhouette Score* for K-means clustering with K value varying from 10 to 80. The scores appear relatively consistent across different K values, predominantly between 0.5 and 0.6. This data shows that K-means can improve the clustering result by increasing the best-performing clustering method, resulting in the highest median *Silhouette Score* of 0.57 for $K = 30$. Additionally, the Inter Quartile Range for K-means clustering tends to be lower, presenting improved variability and consistency. This improvement occurs because K-means clustering separates the data based on the distance of each data to the cluster centroid. The outliers in the K-means cannot cluster non-globular data distribution with inconsistent distance to the centroid, affecting room for improvement for the clustering method.

The third image, **Figure 5(c),** presents *Silhouette* scores for DBSCAN clustering across various e values, determining the minimum distance between points considered part of a cluster. This research utilizes e values ranging from 50 to 450. For the best-performing clustering result ($e = 100$), the *Silhouette* scores show a significant increase in the box plot median, indicating that the cluster is more well-defined than the other clustering method. Furthermore, there is a noticeable variability in the scores for this data. Lastly, **Figure 5(d)** presents a single box plot for Ensemble Clustering. The Ensemble Clustering utilization based on the previous evaluation that K-means can effectively improve the current disaster victim clustering process, especially when faced with a geographical boundary. However, utilizing K-means requires further improvement in several locations with smaller data differences. As a result, leveraging DBSCAN to refine the clustering process by excluding the outlier within the cluster can contribute to a more effective and efficient evacuation strategy.

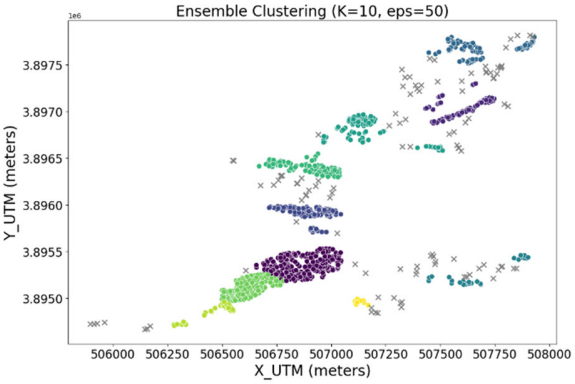
The methodology for determining the optimal combination of K-means and DBSCAN can follow two methods. The first method involves searching all possible combinations of K and e sequentially to find the most effective configuration [67]. However, this method requires extensive resources and may be impractical during time-sensitive disasters. Therefore, the second approach is utilized in this study. The second method is a sequential Ensemble process, assuming that each sequence's best result represents the best Ensemble clustering combination [68]. This method is relevant to the proposed framework because each clustering method serves a specific function in the clustering process and does not affect the other clustering methods. Furthermore, this study demonstrates potential improvements in disaster response by creating a combination of clustering techniques that aid in the operational aspects of emergency management. The Ensemble method's adaptability and efficiency make it suitable for real-time applications, where stakeholders must make immediate decisions based on available data.

As a result, the Ensemble clustering for Nakdong River case in this study is executed with fixed parameters of $K = 30$ and $e = 100$. The *Silhouette Score* of the Ensemble Cluster is slightly higher (0.58) with fewer outliers and IQR, indicating lower variability in the clustering result. This figure shows significant findings suggesting that Ensemble Clustering has generated more cohesive and well-separated clusters for this specific parameter configuration.

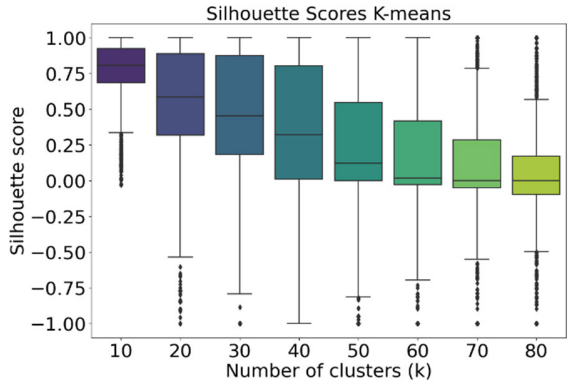
This study extended the analysis to evaluate the *Silhouette Score* across different geographic areas, specifically focusing on the East Coast and Central Canal as additional analysis for to the Nakdong River. Central Canal and East Coast dataset visually recognized has a smaller area data set compared to the Nakdong River Area.

In areas with similar population characteristics, a smaller region is typically related to a lower total population. This variation in total population can influence the clustering result, as illustrated in **Figure 6** and **Figure 7**. For instance, in the K-means clustering algorithm, the Central Canal and East Coast need fewer clusters ($K=10$) than the Nakdong River. This occurs because this location's total population is smaller than the Nakdong River areas. Furthermore, data distribution along the Nakdong River is more decentralized, whereas the Central Canal and East Coast are more centralized. This different data distribution results in higher silhouette scores for the Central Canal and East Coast, providing insight into decision-making. This contribution demonstrates that the proposed framework is suitable for a variety of data distribution types of data and geographical situations. This adaptability further highlights the flexibility of the clustering approaches in identifying relevant clusters across diverse environments. Furthermore, the DBSCAN serves as an outliers detector, determining which epsilon values (e) are most relevant for each area.

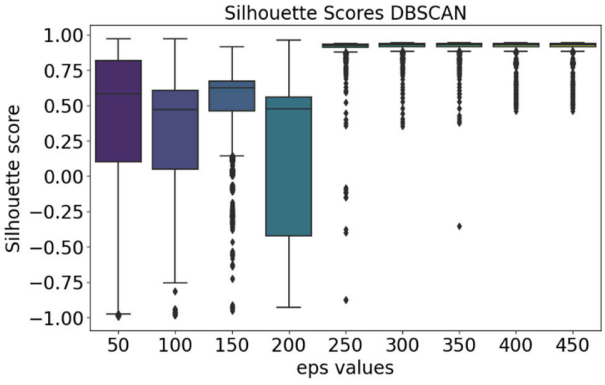
The Central Canal contains more divergent and dense data, which DBSCAN effectively separates by removing unrelated data points from the main clusters (**Figure 6(a)**). In contrast, the East Coast has a more compact data distribution with fewer outlier (**Figure 6(b)**). Following the analysis, the selected e value for the DBSCAN algorithm is determined by considering the highest value from the boxplot analysis. This study, additionally, considers outlier values, which indicate probable anomalies in the *Silhouette Score* distribution [68]. Therefore, $e = 50$ and $e = 100$ are utilized for the Central Canal and the East Coast. As a result, the proposed clustering framework consistently support the goal of increasing *Silhouette Score* while reducing outliers. This consistent application across the Nakdong River, Central Canal, and East Coast demonstrates that the silhouette score can effectively represent well-separated datasets in various clustering scenarios. Using these techniques, the study highlights the importance of the proposed cluster analysis and improves disaster management strategies by providing clear, actionable insights into the spatial distribution of risk and resources.



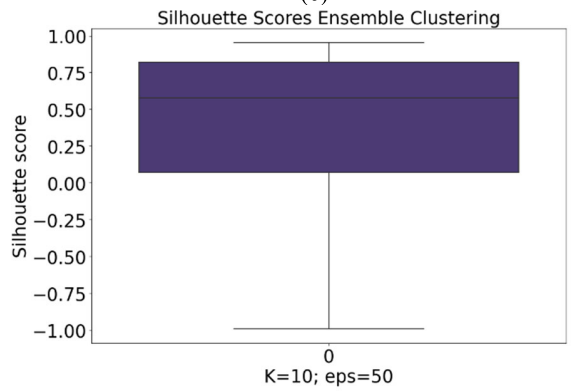
(a)



(b)

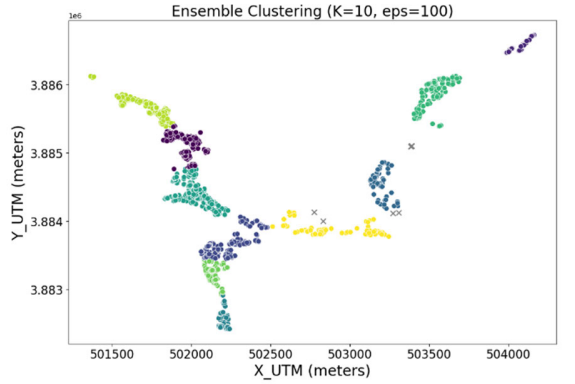


(c)

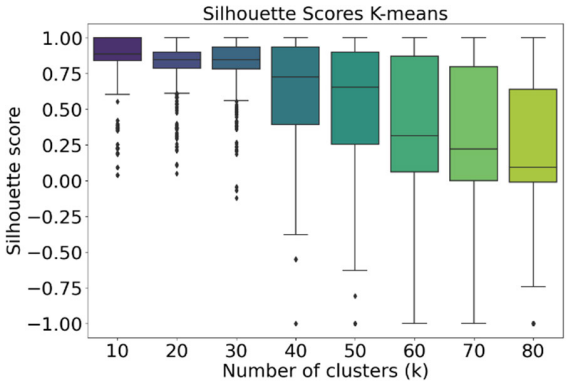


(d)

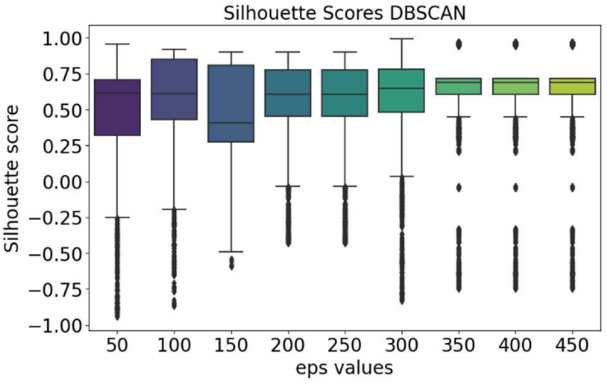
FIGURE 6. Data distribution representation for central canal data set.



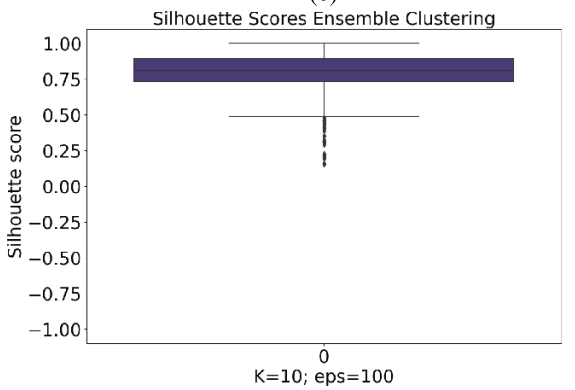
(a)



(b)



(c)



(d)

FIGURE 7. Data distribution representation for east coast data set.

B. VICTIM AGGREGATION IN EVACUATION SIMULATION

The main objective of this research is to modify the disaster victim aggregation method in determining the disaster victim evacuation strategy. Four different clustering methods (*i.e.,* grid-based, K-means, DBSCAN, and Ensemble Clustering) are utilized to cluster the disaster victim. As proof of concept, an evacuation simulation is generated to simulate the evacuation process. Several steps are proceeded to generate real evacuation scenarios. The first step is generating the cluster-shelter couple dataset and assuming each person in the one cluster corresponds to the shelter location. The number of chosen shelters can vary depending on the parameters utilized for each method. This cluster-shelter couple works as an input for the evacuation simulation. Furthermore, this research must carefully transform the cluster-shelter couples into edges-to-edge couples, initially in a coordinate form.

This research used SUMO to model the evacuation process. The simulation utilizes edges-to-edge couples stored in the *passenger.xml* file to represent the evacuee properties and movement characteristics. A snapshot of the evacuation process in the SUMO simulation is shown in **Figure 8**. At the initial time t = 0, disaster victims start evacuating from their original location. Subsequently, the victims continue to evacuate by taking the shortest route to finalize the evacuation process. This simulation dynamically simulates disaster victims during evacuation situations by adjusting disaster victims moving speed and route based on the surrounding environmental conditions. Therefore, evacuees from high-density areas move slower than those from lower-density areas. Furthermore, the distance of individuals to the shelter location emerges as a significant factor influencing the evacuation process [69].

The distance to the designated shelter plays a pivotal role in shaping evacuees evacuation decisions and behaviors. Evacuees located closer to the shelter tend to complete evacuation faster than the evacuees with a longer distance from the shelter. This understanding is crucial because inaccurate clustering of disaster victims may result in assigning them to incorrect locations, potentially increasing the evacuation distances. Therefore, evacuation time result is an essential parameter in evaluating various disaster victim aggregation methods

The comparison of the evacuation time results in the SUMO simulation for various clustering algorithms and clustering parameters is illustrated in **Figure 9**. The *x-axis* presents the evacuation time in minutes, while the *y-axis* represents the proportion of evacuees who have completed the evacuation process. This figure reveals a relevant pattern across most of the evacuation simulation results related to the evacuation process, suggesting the effectiveness of this simulation for developing evacuation strategies. Evaluating evacuation time performance across different aggregation algorithms is crucial for identifying the most effective method. It is important to note that different clustering methods utilize various parameters (*N, K,* and *e*).

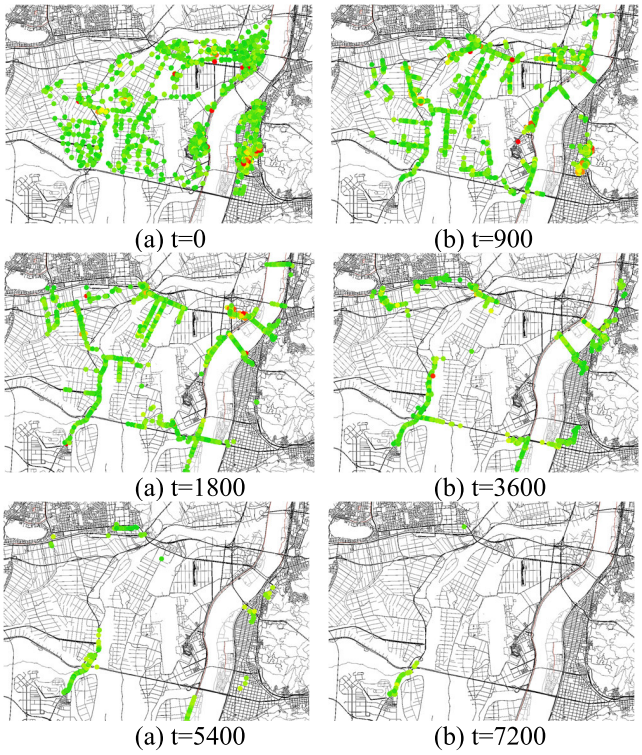
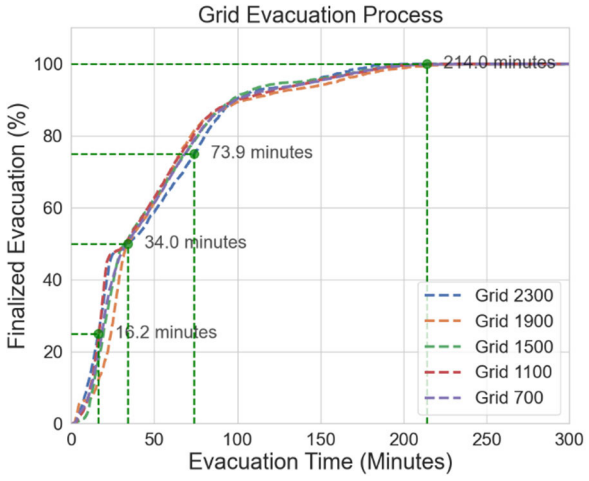
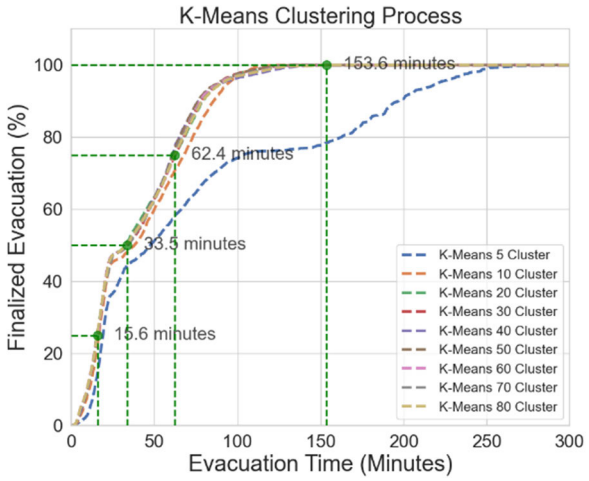


FIGURE 8. Snapshot of evacuation process in SUMO simulation for different time step.

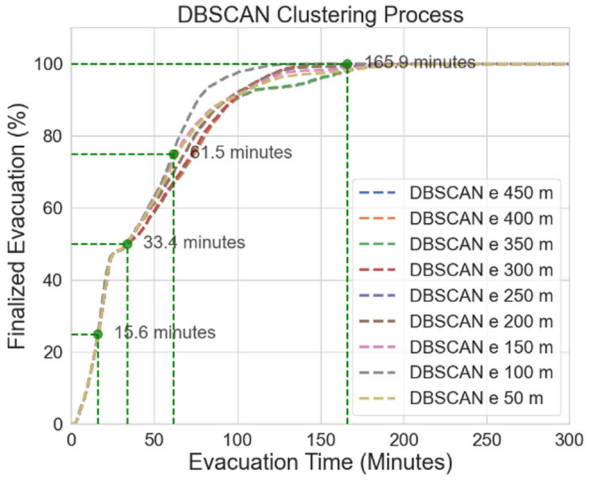
This analysis investigates the influence of different parameter values for each clustering algorithm. Grid-based clustering (**Figure 9(a)**) utilizes four different grid sizes ranging from 700 to 2300 m with intervals of 400 m. The visual assessment indicates that reducing the grid size led to an accelerated evacuation process, especially between the early and mid-evacuation processes (50% evacuation). In the final evacuation stage, a grid size of 2,300 meters results in the fastest evacuation process. While there are slight performance variations among different grid sizes, for the overall evacuation time (100% evacuation), the total evacuation duration consistently hovered around 214 minutes. However, different reference points for the grid-based method across multiple repetitions suggest that the grid-based approach may not be generalizable to all evacuation scenarios. **Figure 9(b)** explores various K values in the K-means clustering. A Lower number of clusters resulted in longer evacuation time. It is relevant to the box plot result that shows high variability in the cluster result. Increasing the number of clusters improves the evacuation times across all evacuation stages. However, the process found a point that the increase in cluster number does not correspond to improvement in evacuation time. This trend indicates that K-means clustering utilization leads to an optimum point in the disaster aggregation for the evacuation process, and the K-value increase does not significantly change the result. K-means clustering also improves the evacuation time by 61 minutes (28.5%) compared to grid-based clustering, potentially leading to the development of life-saving strategies.



(a)



(b)



(c)

FIGURE 9. Evacuation performance for different parameters within aggregation method (a) Grid-based (b) K-Means (c) DBSCAN.

Figure 9(c) provides an evacuation performance from the DBSCAN disaster victim aggregation method. DBSCAN utilizes the e parameter to present the distance to the neighborhood. When comparing the evacuation processes during the early and mid-stages, it is observed that various parameters consistently yield similar results in the evacuation process. Furthermore, the variability starts from the mid-end evacuation process. A lower e value tends to result in a faster evacuation process. The best-performing evacuation process corresponds to $e = 100$, with 165 minutes in completing the evacuation process. Compared with other disaster victim aggregation methods, DBSCAN has a longer evacuation time than the K-Means clustering. However, DBSCAN offers an advantage during the late stages (75% evacuation) of the evacuation process. It can identify and exclude cluster outliers, assigning them to individual cases. In the late stage, the DBSCAN method required 61.5 minutes, slightly less than the 62.4 minutes required by the grid-based and K-means methods. When comparing different parameters in the DBSCAN, it is observed that larger clusters (higher e value) performed similarly to those generated by the grid-based clusters. In comparison, lower e values led to smaller clusters and a faster evacuation process (**Figure 9(c)**). Considering the advantages of late-stage evacuation in DBSCAN clustering and the ability of K-means clustering to provide a faster evacuation time, there is a theoretical possibility of improving the clustering process by combining these two methods.

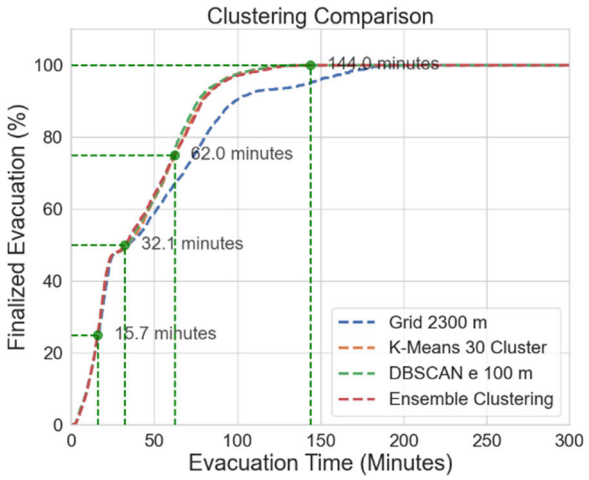


FIGURE 10. Evaluation of evacuation for best performing parameter in different algorithms.

The utilization of Ensemble Clustering can contribute to refining the evacuation process. The implementation of a two-step analysis offers a more cohesive clustering result. This improved clustering has implications for reducing evacuation time. **Figure 10** presents the comparison between various disaster victim aggregation methods. K-means clustering, DBSCAN, and Ensemble Clustering outperform the grid-based clustering method. While K-means, DBSCAN, and Ensemble Clustering exhibit nearly identical evacuation profiles, simulation results indicate that Ensemble Clustering accelerates the total evacuation process by 9 minutes compared to K-means clustering. This phenomenon occurs because DBSCAN refines the cluster formations generated by K-means. The improved aggregation creates more

accurate cluster-shelter pairs for the evacuation process and reduces evacuation time. Furthermore, the applicability of the proposed method utilization is evaluated in another location to ensure that framework generalizability. Therefore, the same clustering value combinations are developed by considering the grid-based, K-means, DBSCAN, and finally assessing the Ensemble method. The evacuation completion time results demonstrate consistency, with the Ensemble method outperforming the other methods (**Table 1**). The grid-based method presents the clustering effectiveness because the Ensemble Clustering takes advantage of both DBSCAN and K-means resulting in the fastest evacuation process.

TABLE 1. Evacuation completion time for different evacuation scenarios and locations.

	Nakdong(s)	C. Canal(s)	E. Coast(s)
Grid-Based	12840	1424	2102
K-Means	9216	858	1641
DBSCAN	9900	891	1190
Ensemble	8640	790	1137

For a more comprehensive assessment, this study carries out an assessment of evacuation at the individual level. The individual-level assessment is essential as each evacuee has a different evacuation profile. Additionally, the most crucial parameter lies in the time required for evacuation. **Figure 11** is a histogram illustrating the improvement in individual evacuation time in the Nakdong River Case. From theAn Ensemble Clustering technique improves the evacuation time for 4,141 evacuees, with the most significant improvement being 42.4 minutes. On average, the evacuation time improvement is 3.21 minutes.

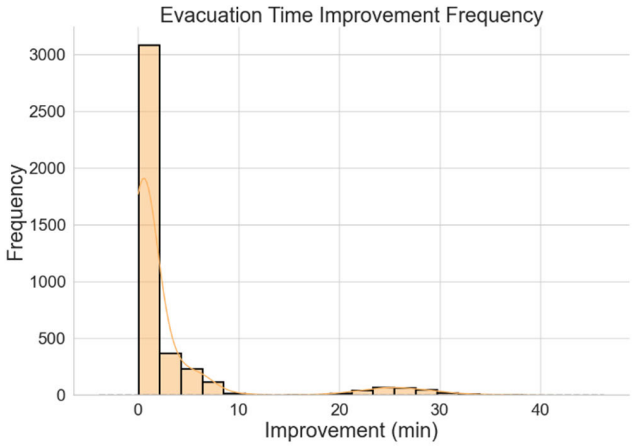


FIGURE 11. Individual evacuation time improvement during evacuation process for (a) Nakdong river, (b) Central canal, (c) East coast.

In the context of clustering applications for the disaster victim aggregation process, a possible exploration has emerged concerning the comparative assessment of various algorithms for an equivalent number of clusters. **Figure 12** presents a comparative analysis of different algorithms with the same number of clusters. For example, for cluster numbers equal to 30, the parameterization that results in a similar number of clusters is the DBSCAN algorithm, configured with an e value of 300 m and a grid-based approach with a grid size of 2000 m. The figure's findings indicate that the number of clusters does not directly represent the optimal parameters across different algorithms.

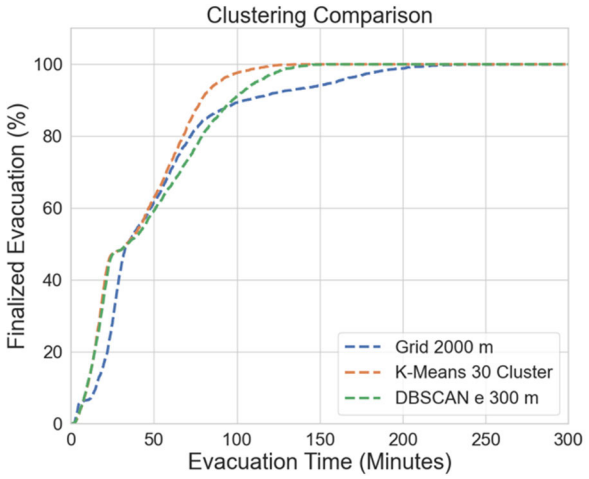


FIGURE 12. Comparison of different methods in 30 cluster evacuations.

This observation underscores an essential aspect for future research: optimizing the evacuation process is unrelated to the number of clusters. Therefore, evaluating the most relevant parameters, such as the number of clusters (K), e, and grid size for each algorithm, is crucial during the initial stages of clustering analysis.

C. INDIVIDUAL WALKING SPEED

The simulation in this study performed an advanced analysis of individual evacuation speed responses to the surrounding environment. The model considers factors such as crowd density which influences dynamic changes in walking speeds based on the situation. **Figure 13** illustrates an example of evacuation speed variation for four individuals, P_1 through P_4, during the first 20 minutes of an evacuation. The graph presents a fluctuating walking speeds that reflect various field conditions. Slower speeds indicate higher densities of people or obstacles. In contrast, higher speeds indicate less pedestrian traffic or the presence of obstacles. This variation demonstrates the capability of this model to simulate realism in the evacuation process and simulating the individual movement complexities during evacuation scenarios.

D. VULNERABLE POPULATION EVALUATION

Population demographics can have a significant impact on evacuation outcomes. To evaluate a more realistic evacuation process, this study conducted an additional analysis to assess the impact on vulnerable populations. The vulnerable

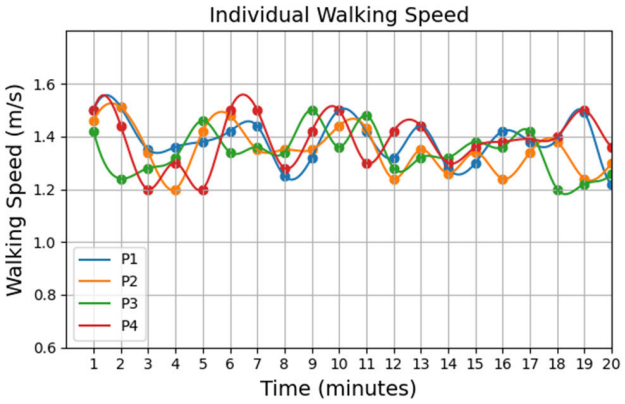


FIGURE 13. Individual walking speed.

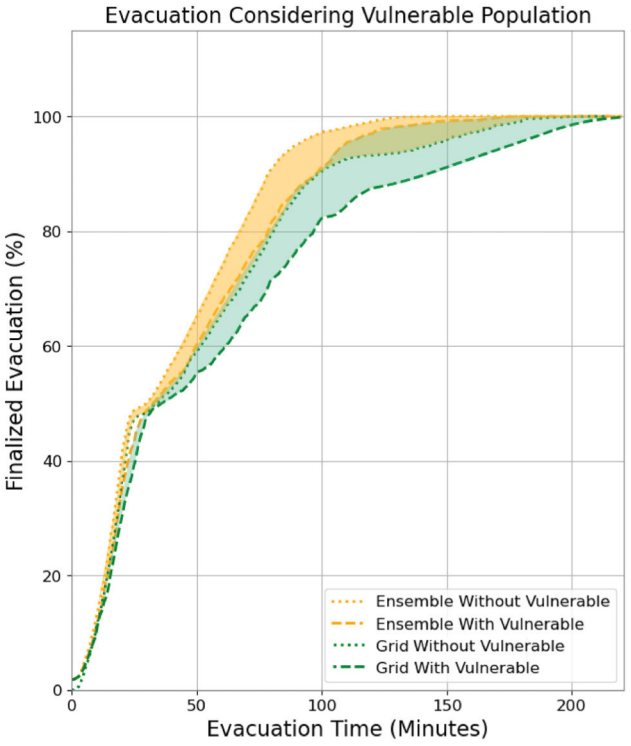
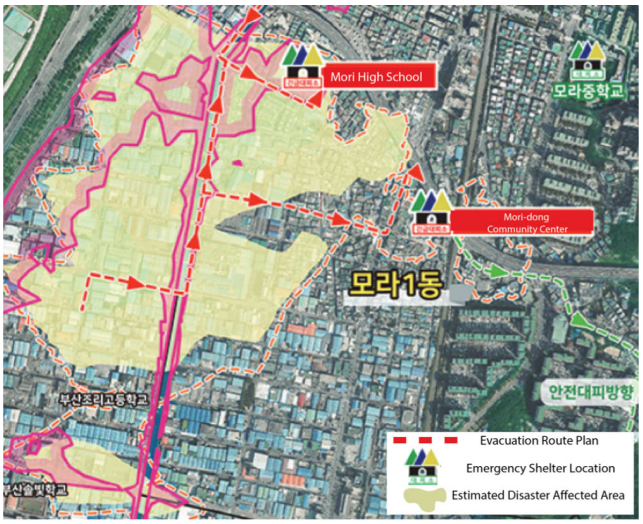
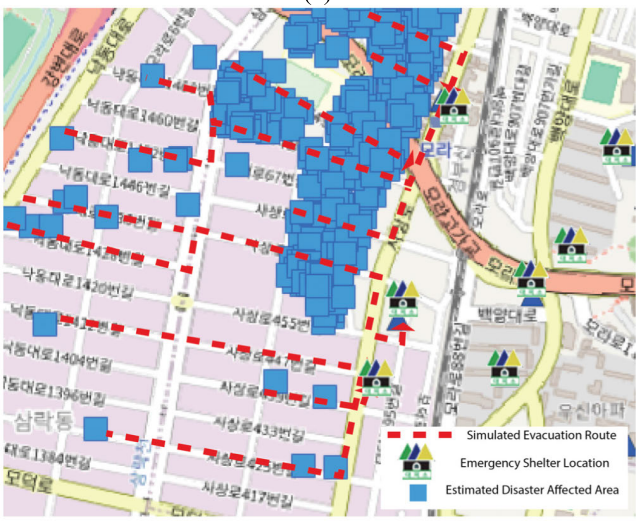


FIGURE 14. Evacuation considering vulnerable population.

population is defined as people over 60 and children under the age of nine. This study modifies the walking speed for the vulnerable population, which has slower evacuation rates, to represent evacuation progress and its temporal implications. **Figure 14** compares evacuation scenarios with and without vulnerable populations in two evacuation aggregation scenarios: Ensemble clustering and grid-based clustering. The dotted line in the figure represents evacuation without vulnerable populations, and the dashed line represents evacuation with vulnerable populations. Additionally, the orange colored line is a scenario using the Ensemble method, and the lined graph uses a grid-based scenario. The analysis demonstrates that incorporating the vulnerable demography increases the evacuation time by approximately 35 minutes



(a)



(b)

FIGURE 15. (a) Busan government evacuation plan (b) Simulated evacuation process.

or equal to 24.3%, especially for the Ensemble clustering method.

The evacuation process is segmented into three phases for a more comprehensive analysis: initial, mid-term, and end phase. During the initial phase, this study observes a rapid increase in the evacuation completion rate of evacuated individuals without any significant difference between the with and without vulnerable populations for Ensemble and grid-based clustering. The mid-term evacuation phase presents the divergence between Ensemble clustering with a vulnerable population and Ensemble clustering without a vulnerable population. For instance, at the 50-minute evacuation time, the Ensemble clustering with the vulnerable population reached 60% and the Ensemble clustering without the vulnerable population resulted in a higher evacuation rate of 63%. This difference can be attributed to the increasing complexities the evacuee encounters, especially the vulnerable population with a slower evacuation rate. Furthermore. The delta increases over time, demonstrating

how the vulnerable population contributes to evacuation. The Ensemble model without vulnerable population finalized the evacuation process when $t = 144$ minutes while the with vulnerable population finalized when $t = 175$ minutes.

E. IMPROVEMENT FROM THE GOVERNMENT EVACUATION SCENARIO

This study focuses on improving existing evacuation scenarios. **Figure 15** presents an example of the evacuation procedure in Mori-Dong, Sasang-Gu, and Busan, South Korea. **Figure 15(a)**, presents the disaster evacuation information provided by the Busan Government, South Korea. This figure provides comprehensive data, including details on disaster risk areas, shelter locations, and evacuation route plans, presenting a static map for the evacuation process. However, this approach faces challenges for different locations and scales of disaster. On the contrary, **Figure 15(b)** illustrates the simulated evacuation process, providing a key advantage of this evacuation framework. This evacuation framework provides an advantage of utilizing multi-source data (*i.e* shelter locations and disaster-affected areas), which enables a comprehensive evaluation of the evacuation process. Multi-source data utilization allows the simulation to identify the nearest emergency shelter locations to the disaster victim and decrease the evacuation process. Moreover, the evacuation simulation offers more detailed evacuation routes, including a number of people passing through the road network. Detailed evacuation routing information provides the stakeholders with more accurate data for allocating evacuation officers and resources effectively. Additionally, the framework facilitates dynamic routing, enabling adaptability to different disaster situations and ensuring optimal response strategies.

IV. CONCLUSION

This research comprehensively evaluates various disaster victim aggregation methods in a city-scale evacuation, aiming to decrease the evacuation time. The presented process is inspired by disaster victim locations that are not equally distributed and affected by surrounding geographical features. Therefore, this study introduces a machine learning algorithm in disaster victim aggregation to enhance the traditional grid-method aggregation. These algorithms are utilized to divide the disaster victims into smaller groups and assign each group to an emergency. Grid-based aggregation is advantageous due to interpretability by stakeholders. However, it is sensitive to the initial parameter change and fails to respond to the geographical variation. The proposed method is optimized by exploring the impact of the internal parameter on the aggregation process. Visual analysis of the clustering and *Silhouette Score* for the main cased study results revealed unique characteristics of each method. The K-means clustering improves the victim aggregation, resulting in a better best-performing *Silhouette Score* from 0.51 to 0.57 compared to the grid-based method. A higher *Silhouette Score* represents better cohesion and separation of the clustered data points. Improved cohesion and separation indicate a more effective disaster victim aggregation.

Consequently, this refinement ensures a more effective evacuation process. The effectiveness of K-means clustering in improving disaster victim evacuation processes is demonstrated through an evacuation simulation. Compared to the traditional grid-based method, K-means significantly reduces evacuation time from 214 minutes to 153.6 minutes, equal to around 61 minutes (28.5%), and potentially corresponds to a life-saving situation. Furthermore, DBSCAN can cluster the data based on the distance to the neighboring data, resulting in a capability to aggregate non-globular data distribution. However, DBSCAN is disadvantageous for a high-density data distribution that generates a relatively big cluster that can decrease the clustering quality. DBSCAN has a higher median *Silhouette Score* with a higher number of outliers. The outliers value indicates inconsistencies in the clustering process. The results from the evacuation model provide insights into the DBSCAN evacuation process effectivity. Despite having a better median performance, DBSCAN has a longer evacuation of 12 minutes compared to the K-means evacuation. DBSCAN still contributes to reducing evacuation time by 49 minutes, or about 23%, compared to the traditional grid-based method. The Ensemble clustering approach significantly optimizes the evacuation process, achieving a reduction of 9 minutes in overall evacuation time when compared to the K-means clustering method. This efficiency establishes the Ensemble method as the quickest strategy for evacuation processes. Moreover, a detailed analysis of individual evacuation times highlights an improved performance of the Ensemble technique. Specifically, the evacuation times of 4,141 evacuees have improved, with the most significant improvement being 42.4 minutes and an average improvement of 3.21 minutes.

Furthermore, additional evaluations to other disaster-prone areas (East Coast and Central Canal) are conducted to address the generalizability of the proposed framework. The results for these new locations present consistency that the Ensemble clustering method can decrease the evacuation time and outperform the other aggregation methods. Additionally, this framework improves the evacuation procedures by offering a dynamic routing and adaptability to various disaster locations and data distribution. These results underscore the practical advantages of employing the Ensemble clustering method in disaster management scenarios. By optimizing evacuation profiles and minimizing individual evacuation times, the Ensemble approach enhances overall evacuation efficiency, potentially mitigating the impact of disasters on affected populations.

REFERENCES

[1] B. Merz, G. Blöschl, S. Vorogushyn, F. Dottori, J. C. J. H. Aerts, P. Bates, M. Bertola, M. Kemter, H. Kreibich, U. Lall, and E. Macdonald, "Causes, impacts and patterns of disastrous river floods," *Nature Rev. Earth Environ.*, vol. 2, no. 9, pp. 592–609, Aug. 2021, doi: 10.1038/s43017-021-00195-3.

[2] R. S. Schumacher, "Heavy rainfall and flash flooding," in *Oxford Research Encyclopedia of Natural Hazard Science*. New York, NY, USA: Oxford Univ. Press, 2017, doi: 10.1093/acrefore/9780199389407.013.132.

[3] Y. Duan, T. Liu, F. Meng, M. Luo, A. Frankl, P. De Maeyer, A. Bao, A. Kurban, and X. Feng, "Inclusion of modified snow melting and flood processes in the SWAT model," *Water*, vol. 10, no. 12, p. 1715, Nov. 2018, doi: 10.3390/w10121715.

[4] W. Shrum, "What caused the flood? Controversy and closure in the Hurricane Katrina disaster," *Social Stud. Sci.*, vol. 44, no. 1, pp. 3–33, Feb. 2014, doi: 10.1177/0306312713498654.

[5] *Impact of the 2010 Pakistan Floods on Rural and Urban Popula6ons at Six Months—PMC* Accessed: May 1, 2024. [Online]. Available: hEps://www.ncbi.nlm.nih.gov/pmc/ar6cles/PMC3441151/

[6] B. -W. Shen, R. Atlas, O. Reale, S. -J. Lin, J. -D. Chern, J. Chang, C. Henze, and J. -L. Li, "Hurricane forecasts with a global mesoscale-resolving model: Preliminary results with Hurricane Katrina (2005)," *Geophys. Res. Lett.*, vol. 33, no. 13, pp. 1–5, Jul. 2006. [Online]. Available: https://agupubs.onlinelibrary.wiley.com/doi/epdf/10.1029/2006GL026143, doi: 10.1029/2006gl026143.

[7] X. Chen and F. B. Zhan, "Agent-based modelling and simulation of urban evacuation: Relative effectiveness of simultaneous and staged evacuation strategies," *J. Oper. Res. Soc.*, vol. 59, no. 1, pp. 25–33, Jan. 2008, doi: 10.1057/palgrave.jors.2602321.

[8] E. Estrada, "What is a complex system, after all?" *Found. Sci.*, pp. 1–28, May 2023. [Online]. Available: https://scholar.google.com/scholar?hl=en&as_sdt=0%2C5&q=What+is+a+Complex+System%2C+Aper+All%3F+estrada&btnG=#d=gs_cit&t=1716499787430&u=%2Fscholar%3Fq%3Dinfo%3Asz-QNTBHsMMJ%3Ascholar.google.com%2F%26output%3Dcite%26scirp%3D0%26hl%3Den, doi: 10.1007/s10699-023-09917-w.

[9] J. C. Chu and C.-Y. Yeh, "Emergency evacuation guidance design for complex building geometries," *J. Infrastruct. Syst.*, vol. 18, no. 4, pp. 288–296, Dec. 2012, doi: 10.1061/(asce)is.1943-555x.0000080.

[10] R. Lovreglio, E. Ronchi, and D. Nilsson, "A model of the decision-making process during pre-evacuation," *Fire Saf. J.*, vol. 78, pp. 168–179, Nov. 2015, doi: 10.1016/j.firesaf.2015.07.001.

[11] G. Bernardini, E. Quagliarini, M. D'Orazio, and M. Brocchini, "Towards the simulation of flood evacuation in urban scenarios: Experiments to estimate human motion speed in floodwaters," *Saf. Sci.*, vol. 123, Mar. 2020, Art. no. 104563, doi: 10.1016/j.ssci.2019.104563.

[12] H. Sbayti and H. S. Mahmassani, "Optimal scheduling of evacuation operations," *Transp. Res. Rec., J. Transp. Res. Board*, vol. 1964, no. 1, pp. 238–246, Jan. 2006, doi: 10.1177/0361198106196400126.

[13] U. Pyakurel and T. N. Dhamala, "Models and algorithms on contraflow evacuation planning network problems," *Int. J. Oper. Res.*, vol. 12, no. 2. Accessed: May 1, 2024. [Online]. Available: https://www.orstw.org.tw/ijor/6_volume12_no2.html

[14] G. J. Lim, M. R. Baharnemati, and S. J. Kim, "An optimization approach for real time evacuation reroute planning," *Ann. Oper. Res.*, vol. 238, nos. 1–2, pp. 375–388, Mar. 2016, doi: 10.1007/s10479-015-2072-5.

[15] U. Pyakurel, H. N. Nath, S. Dempe, and T. N. Dhamala, "Efficient dynamic flow algorithms for evacuation planning problems with partial lane reversal," *Mathematics*, vol. 7, no. 10, p. 993, Oct. 2019, doi: 10.3390/math7100993.

[16] J. Abusalama, S. Razali, Y.-H. Choo, L. Momani, and A. Alkharabsheh, "Dynamic real-time capacity constrained routing algorithm for evacuation planning problem," *Indonesian J. Electr. Eng. Comput. Sci.*, vol. 20, no. 3, p. 1388, Dec. 2020, doi: 10.11591/ijeecs.v20.i3.pp1388-1396.

[17] Y. Ma, X. Liu, F. Huo, and H. Li, "Analysis of cooperation behaviors and crowd dynamics during pedestrian evacuation with group existence," *Sustainability*, vol. 14, no. 9, p. 5278, Apr. 2022, doi: 10.3390/su14095278.

[18] Q. Xiao and J. Li, "Pedestrian evacuation model considering dynamic emotional update in direction perception domain," *Complexity*, vol. 2021, Sep. 2021, Art. no. e5530144, doi: 10.1155/2021/5530144.

[19] M. Liu, W. Zhu, Y. Wang, and J. Zheng, "Modeling and simulation of exit selection behavior in pedestrian evacuation based on information perception and transmission," *Sustainability*, vol. 13, no. 23, p. 13194, Nov. 2021, doi: 10.3390/su132313194.

[20] X. Zhang and G.-L. Chang, "A transit-based evacuation model for metropolitan areas," *J. Public Transp.*, vol. 17, no. 3, pp. 129–147, Sep. 2014, doi: 10.5038/2375-0901.17.3.9.

[21] K. Zhang, R. Su, Y. Lu, and B. Wang, "An optimal traffic dispatching strategy with uncertain evacuees' behaviors under emergency evacuation situation," in *Proc. IEEE 17th Int. Conf. Control Autom. (ICCA)*, Jun. 2022, pp. 746–751, doi: 10.1109/ICCA54724.2022.9831950.

[22] X. Zhao, K. Ji, P. Xu, W.-W. Qian, G. Ren, and X.-N. Shan, "A round-trip bus evacuation model with scheduling and routing planning," *Transp. Res. A, Policy Pract.*, vol. 137, pp. 285–300, Jul. 2020, doi: 10.1016/j.tra.2020.05.008.

[23] X. Li, X. Cui, R. Jiang, and B. Jia, "Bus evacuation during no-notice disasters in downtown areas: A case study of the Zhongguancun area, Beijing," *J. Saf. Sci. Resilience*, vol. 3, no. 3, pp. 235–242, Sep. 2022, doi: 10.1016/j.jnlssr.2022.03.007.

[24] V. C. Pereira and D. R. Bish, "Scheduling and routing for a bus-based evacuation with a constant evacuee arrival rate," *Transp. Sci.*, vol. 49, no. 4, pp. 853–867, Nov. 2015, doi: 10.1287/trsc.2014.0555.

[25] J. Zhang and X. Zhang, "A multi-trip electric bus routing model considering equity during short-notice evacuations," *Transp. Res. D, Transp. Environ.*, vol. 110, Sep. 2022, Art. no. 103397, doi: 10.1016/j.trd.2022.103397.

[26] A.-N. Qazi, Y. Nara, K. Okubo, and H. Kubota, "Demand variations and evacuation route flexibility in short-notice bus-based evacuation planning," *IATSS Res.*, vol. 41, no. 4, pp. 147–152, Dec. 2017, doi: 10.1016/j.iatssr.2017.01.002.

[27] D. Wang and K. Ozbay, "Multi-stage equitable bus-based Hurricane evacuation model with a stochastic driver availability component," *Transp. Res. Rec., J. Transp. Res. Board*, vol. 2677, no. 2, pp. 1619–1632, Feb. 2023, doi: 10.1177/03611981221109158.

[28] M. Gama, B. Santos, and M. Scaparra, "A multi-period shelter location-allocation model with evacuation orders for flood disasters," *EURO J. Comput. Optim.*, vol. 4, nos. 3–4, pp. 299–323, Sep. 2016, doi: 10.1007/s13675-015-0058-3.

[29] S.-K. Wong, Y.-S. Wang, P.-K. Tang, and T.-Y. Tsai, "Optimized evacuation route based on crowd simulation," *Comput. Vis. Media*, vol. 3, no. 3, pp. 243–261, Sep. 2017, doi: 10.1007/s41095-017-0081-9.

[30] N. Nassir, M. Hickman, H. Zheng, and Y.-C. Chiu, "Network flow solution method for optimal evacuation traffic routing and signal control with nonuniform threat," *Transp. Res. Rec., J. Transp. Res. Board*, vol. 2459, no. 1, pp. 54–62, Jan. 2014, doi: 10.3141/2459-07.

[31] D. R. Bish and H. D. Sherali, "Aggregate-level demand management in evacuation planning," *Eur. J. Oper. Res.*, vol. 224, no. 1, pp. 79–92, Jan. 2013, doi: 10.1016/j.ejor.2012.07.036.

[32] Y.-T. Hsu and S. Peeta, "An aggregate approach to model evacuee behavior for no-notice evacuation operations," *Transportation*, vol. 40, no. 3, pp. 671–696, May 2013, doi: 10.1007/s11116-012-9440-7.

[33] M. Asadi and J. Karami, "Modeling of the city evacuation plan in case of earthquake with particle swarm optimization (PSO) and imperialist competition algorithm (ICA)," *Int. J. Disaster Resilience Built Environ.*, vol. 11, no. 1, pp. 134–151, Nov. 2019, doi: 10.1108/ijdrbe-12-2018-0053.

[34] M. J. Alam, M. A. Habib, and D. Husk, "Evacuation planning for persons with mobility needs: A combined optimization and traffic microsimulation modelling approach," *Int. J. Disaster Risk Reduction*, vol. 80, Oct. 2022, Art. no. 103164, doi: 10.1016/j.ijdrr.2022.103164.

[35] X. Zhao, J. Chen, W. Xu, S. Lou, P. Du, H. Yuan, and K. P. Ip, "A three-stage hierarchical model for an earthquake shelter location-allocation problem: Case study of Chaoyang district, Beijing, China," *Sustainability*, vol. 11, no. 17, p. 4561, Aug. 2019, doi: 10.3390/su11174561.

[36] I. Lee, "Geospatial clustering in data-rich environments: Features and issues," in *Knowledge-Based Intelligent Information and Engineering Systems* (Lecture Notes in Computer Science), vol. 3684, R. Khosla, R. J. Howlett, and L. C. Jain, Eds. Berlin, Germany: Springer, 2005, pp. 336–342, doi: 10.1007/11554028_47.

[37] Z. Chen, X. Chen, Q. Li, and J. Chen, "The temporal hierarchy of shelters: A hierarchical location model for earthquake-shelter planning," *Int. J. Geographical Inf. Sci.*, vol. 27, no. 8, pp. 1612–1630, Aug. 2013, doi: 10.1080/13658816.2013.763944.

[38] Y. Li, H. Liu, G.-P. Liu, L. Li, P. Moore, and B. Hu, "A grouping method based on grid density and relationship for crowd evacuation simulation," *Phys. A, Stat. Mech. Appl.*, vol. 473, pp. 319–336, May 2017, doi: 10.1016/j.physa.2017.01.008.

[39] H. Liu, Y. Li, W. Li, D. Lu, and G. Zhang, "A grouping approach based on non-uniform binary grid partitioning for crowd evacuation simulation," *Concurrency Comput., Pract. Exper.*, vol. 31, no. 23, p. e4493, Dec. 2019, doi: 10.1002/cpe.4493.

[40] K. Wang, W. Yuan, and Y. Yao, "Path optimization for mass emergency evacuation based on an integrated model," *J. Building Eng.*, vol. 68, Jun. 2023, Art. no. 106112, doi: 10.1016/j.jobe.2023.106112.

[41] P. Murray-Tuite and B. Wolshon, "Evacuation transportation modeling: An overview of research, development, and practice," *Transp. Res. C, Emerg. Technol.*, vol. 27, pp. 25–45, Feb. 2013, doi: 10.1016/j.trc.2012.11.005.

[42] J. Joo, N. Kim, R. A. Wysk, L. Rothrock, Y.-J. Son, Y.-G. Oh, and S. Lee, "Agent-based simulation of affordance-based human behaviors in emergency evacuation," *Simul. Model. Pract. Theory*, vol. 32, pp. 99–115, Mar. 2013, doi: 10.1016/j.simpat.2012.12.007.
[43] P. M. Ejercito, K. G. E. Nebrija, R. P. Feria, and L. L. Lara-Figueroa, "Traffic simulation software review," in *Proc. 8th Int. Conf. Inf., Intell., Syst. Appl. (IISA)*, Aug. 2017, pp. 1–4, doi: 10.1109/IISA.2017.8316415.
[44] X. Liao, X. Zhao, Z. Wang, K. Han, P. Tiwari, M. J. Barth, and G. Wu, "Game theory-based ramp merging for mixed traffic with unity-SUMO co-simulation," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 52, no. 9, pp. 5746–5757, Sep. 2022, doi: 10.1109/TSMC.2021.3131431.
[45] M. Behrisch, L. Bieker, J. Erdmann, and D. Krajzewicz, "SUMO—Simulation of urban mobility: An overview," in *Proc. 3rd Int. Conf. Adv. Syst. Simulation*, R. I.-R. T. P. Diglio A. Simoni, and R. I.-R. T. P. Georgiy Bobashev, Eds., Oct. 2011, pp. 63–68. Accessed: Jun. 19, 2023. [Online]. Available: http://www.thinkmind.org/index.php?view=instance&instance=SIMUL+2011 and https://sumo.dlr.de/pdf/simul_2011_3_40_50150.pdf
[46] A. Diallo, G. Lozenguez, A. Doniec, and R. Mandiau, "Comparative evaluation of road traffic simulators based on modeler's specifications: An application to intermodal mobility behaviors," in *Proc. 13th Int. Conf. Agents Artif. Intell.*, 2021, pp. 265–272, doi: 10.5220/0010238302650272.
[47] H. Nam, H. Nam, and D. Lee, "Potential of hydrogen replacement in natural-gas-powered fuel cells in Busan, South Korea based on the 2050 clean energy master plan of Busan metropolitan city," *Energy*, vol. 221, Apr. 2021, Art. no. 119783, doi: 10.1016/j.energy.2021.119783.
[48] J.-S. Kim, S.-K. Yoon, M. Choi, and Y.-I. Moon, "A case study of regional risk assessment of river restoration projects: Nakdong river basin, South Korea," *J. Water Climate Change*, vol. 6, no. 3, pp. 628–637, Feb. 2015, doi: 10.2166/wcc.2015.113.
[49] K. Lwin and Y. Murayama, "A GIS approach to estimation of building population for micro-spatial analysis," *Trans. GIS*, vol. 13, no. 4, pp. 401–414, Aug. 2009, doi: 10.1111/j.1467-9671.2009.01171.x.
[50] S. Geng, H. Hou, and S. Zhang, "Multi-criteria location model of emergency shelters in humanitarian logistics," *Sustainability*, vol. 12, no. 5, p. 1759, Feb. 2020, doi: 10.3390/su12051759.
[51] Ministry of the Interior and Safety. *National Disaster and Safety Portal*. Accessed: Feb. 16, 2024. [Online]. Available: https://www.safekorea.go.kr/idsiSFK/neo/sfk/cs/contents/civil_defense/
[52] A. Kusari, P. Li, H. Yang, N. Punshi, M. Rasulis, S. Bogard, and D. J. LeBlanc, "Enhancing SUMO simulator for simulation based testing and validation of autonomous vehicles," in *Proc. IEEE Intell. Vehicles Symp. (IV)*, Jun. 2022, pp. 829–835, doi: 10.1109/IV51971.2022.9827241.
[53] A. Likas, N. Vlassis, and J. J. Verbeek, "The global k-means clustering algorithm," *Pattern Recognit.*, vol. 36, no. 2, pp. 451–461, Feb. 2003, doi: 10.1016/s0031-3203(02)00060-2.
[54] W.-L. Zhao, C.-H. Deng, and C.-W. Ngo, "k-means: A revisit," *Neurocomputing*, vol. 291, pp. 195–206, May 2018, doi: 10.1016/j.neucom.2018.02.072.
[55] T. Boonchoo, X. Ao, Y. Liu, W. Zhao, F. Zhuang, and Q. He, "Grid-based DBSCAN: Indexing and inference," *Pattern Recognit.*, vol. 90, pp. 271–284, Jun. 2019, doi: 10.1016/j.patcog.2019.01.034.
[56] B. Leite, P. Azevedo, R. Leixo, and R. J. F. Rossetti, "Simulating a three-lane roundabout using SUMO," in *Intelligent Transport Systems. From Research and Development To the Market Uptake* (Lecture Notes of the Institute for Computer Sciences, Social Informatics and Telecommunications Engineering), A. L. Martins, J. C. Ferreira, and A. Kocian, Eds. Cham, Switzerland: Springer, 2020, pp. 18–31, doi: 10.1007/978-3-030-38822-5_2.
[57] C.-F. Tsai and S.-C. Huang, "An effective and efficient grid-based data clustering algorithm using intuitive neighbor relationship for data mining," in *Proc. Int. Conf. Mach. Learn. Cybern. (ICMLC)*, vol. 2. Guangzhou, China: IEEE, Jul. 2015, pp. 478–483, doi: 10.1109/ICMLC.2015.7340603.
[58] W. Cheng, W. Wang, and S. Batista, "Grid-based clustering," in *Data Clustering*. Boca Raton, FL, USA: CRC Press, 2018, pp. 128–148.
[59] M. Lv, "Application of an K-means improved clustering analysis algorithm in the design of resource management information system," in *Proc. World Autom. Congr. (WAC)*, Oct. 2022, pp. 158–162, doi: 10.23919/WAC55640.2022.9934387.
[60] P. Dubey and A. Rajavat, "Effective K-means clustering algorithm for efficient data mining," in *Proc. 2nd Int. Conf. Vis. Towards Emerg. Trends Commun. Netw. Technol. (ViTECoN)*, May 2023, pp. 1–6, doi: 10.1109/ViTECoN58111.2023.10157179.
[61] D. S. Kim, "Finding the number of clusters using a small training sequence," *IEEE Access*, vol. 11, pp. 25932–25940, 2023, doi: 10.1109/ACCESS.2023.3257163.
[62] H. K. Kanagala and V. V. Jaya Rama Krishnaiah, "A comparative study of K-means, DBSCAN and OPTICS," in *Proc. Int. Conf. Comput. Commun. Informat. (ICCCI)*, Jan. 2016, pp. 1–6, doi: 10.1109/ICCCI.2016.7479923.
[63] J. M. Jobe and M. Pokojovy, "A multistep, cluster-based multivariate chart for retrospective monitoring of individuals," *J. Quality Technol.*, vol. 41, no. 4, pp. 323–339, Oct. 2009, doi: 10.1080/00224065.2009.11917789.
[64] M. Rossi and S. Brinkkemper, "Metrics in method engineering," in *Advanced Information Systems Engineering* (Lecture Notes in Computer Science), J. Iivari, K. Lyytinen, and M. Rossi, Eds. Berlin, Germany: Springer, 1995, pp. 200–216, doi: 10.1007/3-540-59498-1_247.
[65] K. R. Shahapure and C. Nicholas, "Cluster quality analysis using silhouette score," in *Proc. IEEE 7th Int. Conf. Data Sci. Adv. Analytics (DSAA)*, Oct. 2020, pp. 747–748, doi: 10.1109/DSAA49011.2020.00096.
[66] T. Long, *Data Mining: Concepts and Techniques*, 3rd ed. San Mateo, CA, USA: Morgan Kaufmann, 2022, pp. 1–740.
[67] S. Yuan, K. De Roover, and K. Van Deun, "Simultaneous clustering and variable selection: A novel algorithm and model selection procedure," *Behav. Res. Methods*, vol. 55, no. 5, pp. 2157–2174, Sep. 2022, doi: 10.3758/s13428-022-01795-7.
[68] C. H. Sim, F. F. Gan, and T. C. Chang, "Outlier labeling with boxplot procedures," *J. Amer. Stat. Assoc.*, vol. 100, no. 470, pp. 642–652, Jun. 2005, doi: 10.1198/016214504000001466.
[69] M. J. Alam, M. A. Habib, and E. Pothier, "Shelter locations in evacuation: A multiple criteria evaluation combined with flood risk and traffic microsimulation modeling," *Int. J. Disaster Risk Reduction*, vol. 53, Feb. 2021, Art. no. 102016, doi: 10.1016/j.ijdrr.2020.102016.



REZA ASRIANDI EKAPUTRA is currently pursuing the Ph.D. degree with the Department of ICT Integrated Safe Ocean Smart Cities Engineering, Dong-A University. He is a dedicated Researcher whose work is centered around a diverse set of interests in transportation and disaster management. His research interests include transportation modeling, disaster evacuation strategies, machine learning applications, and remote sensing technologies. His contributions in these areas aim to enhance the efficiency and effectiveness of transportation networks while also addressing disaster preparedness and evacuation challenges.



SEONG-HOON KEE is currently the Associate Director and an Associate Professor with the Department of Architectural Engineering and the Department of ICT Integrated Safe Ocean Smart Cities Engineering. His expertise lies in the civil infrastructure systems, where he specializes in the structures condition assessment. Additionally, he is actively involved in the development of non-destructive evaluation systems and demonstrating his commitment to innovative approaches for ensuring the integrity and longevity of infrastructure.



JURNG-JAE YEE holds the positions of the Director and a Professor with the Department of Architectural Engineering and the Department of ICT Integrated Safe Ocean Smart Cities Engineering. His research interests include urban disaster prevention and indoor air quality. Additionally, his work encompasses urban environment planning and evaluation, focusing on creating sustainable and adaptive urban landscapes. His expertise extends to the application of computational fluid dynamics (CFD) in engineering, further contributing to the advancement of technology in urban contexts.